Do We Train on Test Cells?

A Preregistered Audit of Hierarchical Leakage in Cervical Cytology Benchmarks

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Abstract

Public cervical cytology benchmarks release many cell crops cut from far fewer microscope fields or slide images, and are usually split at the crop level, so test cells share their acquisition with training cells. A first study showed that this structure is pervasive and that the shared parent is recognisable, but six per-architecture tests could not show that it inflates accuracy. In a second, preregistered study (1,341 models) we test the question once, with a mixed model sized by simulation: across twelve architectures and twenty new seeds, crop-level splitting raised SIPaKMeD accuracy by 0.97 percentage points (95% CI 0.59–1.39), so parent-level splitting reveals 1.34 times the error rate. On CRIC, whose slide images hold many cells of several classes, the inflation was 12.08 points. A factorial that holds crop and field counts fixed shows that the inflation is leakage (1.73 points) and not the extra training diversity crop-level splits bring (diversity effect 0.00 points). Eight further registered experiments (1,580 runs) locate the leaked signal by elimination. It needs no background and, in an exploratory split, does not depend on shared pixels; it is not a slide-level label prior; and it is not carried by linearly decodable field identity: erasing or adversarially suppressing that identity cut probe recoverability by up to 91% without moving the gap. It is instead tied to how a field renders its cells: matching a test cell's colour statistics to an unseen field removed most of it, although colour augmentation during training did not prevent it. Grad-CAM maps of leaking and non-leaking models were indistinguishable. Neither probes, saliency maps, nor augmentation can certify an evaluation; parent-disjoint splits can. We release the provenance maps, splits, run records, and a risk calculator that predicts a benchmark's exposure from its crop-to-parent table.

1 Introduction

Public cervical cytology datasets are usually released as cell crops, and most of those crops are cut from a much smaller number of microscope fields or slide images. When a benchmark is split at the crop level, crops from one field can land on both sides of the train–test boundary. File-level disjointness then looks like independence while the test set shares stain, illumination, focus, background, and sometimes literal pixels with the training set.

This paper separates three claims that are easy to conflate: *structural non-independence* (test observations share a parent with training observations), *shortcut availability* (the shared parent leaves a learnable fingerprint), and *consequence* (classifiers score higher because they use it). Our first study (Study 1) established the first two for four public datasets: random crop splitting placed a training parent behind 92.6% of SIPaKMeD and 99.97% of CRIC test cells, more than half of SIPaKMeD test crops shared source pixels with a training crop, and a 424-way parent probe reached 109 times chance (79 times with the cell inpainted away), while the one-cell-per-image Herlev control stayed at exactly zero. It did not establish the third. All six architectures scored higher under crop-level than parent-level splitting, by 0.72–2.25 percentage points, yet none of the six per-architecture tests survived Benjamini–Hochberg correction.

That design had four weaknesses, and this paper is a second, separately preregistered study (Study 2) built to remove them:

1. **Specification.** One question was asked six times. We replace the six tests with a single mixed-effects test of the protocol effect, sized by simulation from the Study 1 pilot (twelve architectures, twenty new seeds; Section 5.3).

2. **Availability versus use.** We test whether classifiers *use* acquisition context with counterfactual background transplants (Section 5.5) and whether the inflation tracks the fingerprint when an adversary suppresses it (Section 5.6).

3. **Leakage versus diversity.** At equal crop counts, crop-level training sets contain about 18% more distinct parents than parent-level ones, so the protocol contrast also changes training diversity. A leakage-by-diversity factorial over one common test set separates the two (Section 5.4).

4. **One rescored benchmark.** We rescore CRIC, whose cells are drawn from 400 multi-class slide images (Section 5.7).

Every Study 2 split, model, seed, endpoint, and test was committed before the first Study 2 model was trained. All results are reported, including any that do not support our hypotheses.

2 Related work

Leakage and duplicates. Kapoor and Narayanan catalogue leakage as a cause of irreproducible machine-learning science across fields [1]. Barz and Denzler showed that near-duplicates of CIFAR test images inflate reported accuracy and released a purged test set [2]; Laroca et al. found the same for license plates [3]. In medical imaging, patient identity is recoverable from chest radiographs [4], so image-level splits of multi-image patients can share identity across the boundary. Our setting is a hierarchy rather than duplication: sibling crops are different cells from one acquisition unit.

Shortcut learning. Networks often solve a task through cues that are predictive in the training distribution but unrelated to the intended concept [5]. Medical examples include hospital and scanner signatures in pneumonia detection [6], dataset-source artifacts in COVID-19 radiograph models [7], and models that predict dietary habits from knee radiographs [8]; see also the methodological review of Varoquaux and Cheplygina [9]. Xiao et al. measured background reliance by recombining ImageNet foregrounds with backgrounds from other classes [10]; our transplant arm adapts that design to acquisition context in cytology.

Removing nuisance information. Gradient reversal trains a representation that is uninformative about a nuisance variable by adversarially training a nuisance predictor [11, 12]. We use it as an intervention on the parent fingerprint rather than for domain adaptation.

Variance and inference in benchmarks. Benchmark comparisons inherit variance from data splits and initialization [13]. Crossed mixed-effects models with maximal random-effects structure [14, 15] and Satterthwaite or Kenward–Roger degrees of freedom [16–18] are the standard way to generalise a fixed effect over both items and conditions; simulation-based power analysis handles their non-closed-form power [19].

3 Datasets and provenance

Table 1 lists the four audited datasets and their independent units. SIPaKMeD crops come from cluster fields [20]; Cx22 regions come from CCEDD source images [21]; CRIC cells come from 400 conventional Pap smear images [22]; Herlev releases one cell per image and serves as

Table 1: Audited datasets and their independent units.

Dataset	Observation	Parent	Obs.	Parents	Obs./parent	Released task
SIPaKMeD	cell crop	cluster field	4,049	966	4.19	5-class classification
Cx22	generated ROI	CCEDD source image	1,320	514	2.57	segmentation
CRIC	classified cell	slide image	11,534	400	28.84	6-class classification
Herlev	single-cell image	the image itself	917	917	1.00	7-class classification

Table 2: Structural leakage of a random 80:20 crop split. Straddle and test-crop leakage are percentages (observed: mean ± SD over five seeds); siblings, diversity ratio, and class–parent dependence are closed-form expectations (`cytofair risk-report`); literal overlap is the percentage of test observations sharing source pixels with a training observation (same five seeds). P2 and P3 are exactly zero on every column.

Dataset	Analytic	Observed	Test w/ train parent	Siblings	Diversity	Class dep.	Literal
SIPaKMeD	44.8	45.63 ± 1.04	92.62 ± 0.40	5.2	1.18	1.00	52.62 ± 1.97
Cx22	33.9	34.01 ± 1.10	85.23 ± 3.51	1.9	1.16	–	34.02 ± 1.80
CRIC	94.9	94.35 ± 1.10	99.97 ± 0.08	35.6	1.25	0.68	71.54 ± 0.90
Herlev	0.0	0.00 ± 0.00	0.00 ± 0.00	0.0	1.00	–	0.00

the negative control [23]. For SIPaKMeD, two independent recovery branches (filename parsing, and template matching of each crop against every same-class field) agree on all 4,049 crops, and every recovered box lies inside its field. Raw images are never redistributed; the released artifacts are identifiers, coordinates, hashes, and split assignments.

4 Structural leakage and the parent fingerprint

For a parent with k crops allocated independently with test fraction q, the probability that it straddles the split is $1 - (1-q)^k - q^k$, a test crop has a training sibling with probability $1 - q^{k-1}$, and it has $(1-q)(k-1)$ training siblings on average. Averaging these closed forms over the recovered parents predicts the structural leakage of a random crop split before any split is drawn (Table 2); the observed values agree within Monte Carlo error. The same closed forms give the *training-diversity ratio* between crop-level and parent-level splits of equal size, $\sum_j \{1 - q^{k_j}\}/\{(1-q)m\}$ for m parents, which is 1.18 for SIPaKMeD and 1.25 for CRIC. The `cytofair risk-report` command computes all of these from any crop-to-parent table.

The fingerprint is learnable. A ResNet-18 trained to identify 424 SIPaKMeD fields from two crops each recognised held-out sibling crops at 25.7% accuracy (chance 0.24%), still reached 18.7% from inpainted, cell-removed backgrounds, and fell to chance under shuffled labels (Study 1). In SIPaKMeD every field belongs to one class, so recognising the field implies the label. In CRIC, slide identity explains 1.23 of the 1.82 bits of class entropy (68%).

5 Study 2 design

5.1 Governance

Study 2 was registered as an amendment to the original plan (`PREREGISTRATION_STUDY2.md`, commit `a62e8e7`, 2026-10-18 05:07 IST). Before registration, every split, design table, transplant plan, and input cache was generated and hashed, and one-epoch smoke tests were run only on crop-level splits, so no protocol contrast could be observed. Study 1 runs serve only as pilot estimates; Study 2 seeds are disjoint from Study 1 seeds. While validating the analysis code we fitted the registered model to the Study 1 runs; that post hoc reanalysis is reported in Appendix A and is not evidence for Study 2.

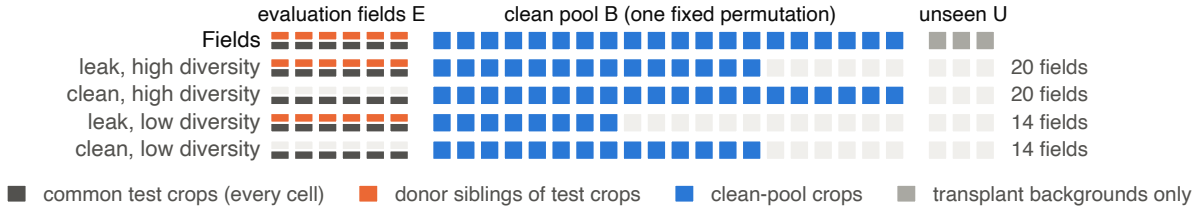


Figure 1: The 2C factorial for one seed. Every cell shares the same test crops (half of each evaluation field's crops). Leak cells add the evaluation fields' remaining sibling crops; clean cells use other fields instead. Within a diversity level, crop and field counts are identical; low-diversity field sets are nested inside high-diversity ones.

5.2 Common training recipe

Every run fine-tunes an ImageNet-initialised backbone with a new linear head: AdamW at 10^{-4} with weight decay 10^{-4}, batch 64, 20 epochs, final-epoch evaluation, no validation split, early stopping, or checkpoint selection; random resized crops (scale 0.8–1.0) with horizontal and vertical flips. Twelve backbones span classic and modern designs: ResNet-18 and -50 [24], DenseNet-121 [25], MobileNet-V2 [26], EfficientNet-B0 [27], ViT-B/16 [28], ConvNeXt-T [29], Swin-T [30], RegNetY-1.6GF [31], EfficientNetV2-S [32], MaxViT-T [33], and the self-supervised DINOv2 ViT-S/14 [34]. Runs are deterministic (two identical runs gave bitwise-identical logits). Each run keeps its per-crop test logits, its final weights, and penultimate features of a fixed set of probe crops, so confusion matrices and post hoc analyses need no retraining.

5.3 H1: the protocol effect, tested once

Twelve architectures are trained on P1 (stratified crop-level 80:20), P2 (parent-disjoint), and P3 (parent-disjoint after merging near-duplicate-linked parents) for twenty new seeds: 720 runs. P1-prime, the size-matched control, is byte-identical to P1 for every seed because P2 always has exactly 810 test crops. The primary endpoint is test accuracy, analysed with

$$\text{acc} \sim x + (1 + x \mid \text{architecture}) + (1 \mid \text{seed}) + (1 \mid \text{split}), \tag{1}$$

where $x = \pm\frac{1}{2}$ codes P1 versus P2, so the fixed effect is the P1–P2 difference averaged over architectures, and *split* is the seed-by-protocol partition shared by all twelve architectures. Omitting it would count twelve evaluations of one test set as twelve independent test sets. We fit by REML with lme4 [15] and test with Satterthwaite degrees of freedom [18], two-sided $\alpha = 0.05$; H1 is supported if $p < 0.05$ and the estimate is positive. The design was chosen by simulating (1) from the Study 1 variance components: twenty seeds and twelve architectures give 0.98 power for a 0.8-pp effect, and 0.78 with every variance inflated by half (Appendix B). Kenward–Roger degrees of freedom, a binomial mixed model, a two-way cluster bootstrap, and a seed-level sign-flip test are reported as sensitivity analyses; per-architecture differences are descriptive.

5.4 Leakage versus diversity at matched size (H3)

The 2C factorial (Figure 1) fixes one test set per seed and varies only the training set. Thirty percent of SIPaKMeD fields with at least two crops are evaluation fields; half of each one's crops (rounded down) form the common test set (about 527 crops) and the rest are its *donor siblings*. Six fields per class are reserved as unseen background donors for Section 5.5. Four training sets each hold exactly 220 crops per class. The clean cells draw only from non-evaluation fields; the leak cells replace the last $|E|$ fields of the same random permutation with the evaluation fields' donor siblings. High-diversity cells use 720 distinct fields and low-diversity cells 504, a 1.43-fold contrast that brackets the natural 1.18. Within a diversity level, leak and clean therefore have the same crop count, the same number of distinct fields, and the same allocation rule; they differ

only in whether the test fields' siblings are present. ResNet-50, DenseNet-121, ConvNeXt-T, and ViT-B/16 are trained on all four cells for ten seeds (160 runs). H3 is the leak main effect in

$$\text{acc} \sim \text{leak} \times \text{div} + (1 + \text{leak} \mid \text{architecture}) + (1 \mid \text{seed}) + (1 \mid \text{training set}). \tag{2}$$

Because the low-diversity leak cell gives each test crop more training siblings (3.0 versus 2.0), the diversity effect is estimated within clean cells only and the interaction is exploratory.

5.5 Counterfactual background transplants (H4)

Every 2C model is also evaluated on composites of its test set. Each test cell's foreground (the Study 1 background-only mask) is feathered onto the inpainted background of a donor crop: its own background (*self*, which isolates seam and inpainting artifacts), a sibling from the same field (*sibling*), an unseen field of the same class (*unseen same-class*), and unseen fields of every other class. Leak models have seen each test cell's field context with the correct label; clean models have not. The per-run quantity route = acc(sibling) − acc(unseen same-class) measures how much the cell's own acquisition context contributes beyond class-typical context, and H4 is its leak-minus-clean difference:

$$\text{route} \sim \text{leak} + \text{div} + (1 + \text{leak} \mid \text{architecture}) + (1 \mid \text{seed}) + (1 \mid \text{training set}). \tag{3}$$

The ratio of H4 to the leakage effect on unmodified images estimates the share of the inflation carried by the context route. The probability shift toward the donor class under other-class backgrounds measures, separately, whether classifiers read class information from backgrounds at all.

5.6 Adversarial fingerprint suppression (H5)

A two-layer parent-identity head is attached to the penultimate features through a gradient-reversal layer [11, 12], with the DANN ramp reaching a maximum coefficient $\lambda \in \{0, 0.01, 0.03, 0.1, 0.3, 1, 3\}$. The head only discriminates fields *within* the sample's class, so suppressing it never requires removing class information. ResNet-50 and ConvNeXt-T are trained on the natural P1 and P2 splits of ten seeds (280 runs). Fingerprint recoverability R is the within-class field-identification accuracy of a logistic-regression probe on frozen features of the balanced 424-field probe crops, reported against its class-only floor. Within the non-degenerate regime (mean P2 accuracy within 1 pp of $\lambda = 0$), and provided suppression removes at least a quarter of the above-floor recoverability, H5 is a positive slope of the per-seed P1–P2 gap on R. If the shortcut account is right, the gap closes as the fingerprint is suppressed while P2 accuracy stays flat; if P1 stays inflated while R collapses, the gap is something else.

5.7 CRIC (H2)

CRIC cells are 100 × 100 windows centred on the published nucleus coordinates of the 400 Figshare slide images (six Bethesda classes). ResNet-18, ResNet-50, DenseNet-121, EfficientNet-B0, ConvNeXt-T, and ViT-B/16 are trained on P1, P1-prime (which differs from P1 by 2–11 training cells), and P2 for ten seeds (180 runs). H2 applies (1) to CRIC accuracy.

5.8 Multiplicity

H1 is the single primary endpoint. H2–H5 form one Holm family [35] (family-wise $\alpha = 0.05$); each is supported only if its adjusted $p < 0.05$ and its estimate has the predicted sign. All other analyses are secondary or exploratory.

Table 3: Registered Study 2 tests. Estimates in percentage points (H5: pp of gap per pp of recoverability). H1 is tested alone; H2–H5 are one Holm family. H1's primary inference is the two-way cluster bootstrap because the registered residual check flagged heavy tails.

	Question	Estimate	95% CI	p	Verdict
H1	SIPaKMeD, P1 − P2 accuracy	0.97	0.59 to 1.39	<0.001	supported
H2	CRIC, P1 − P2 accuracy	12.08	11.16 to 12.99	<0.001	supported
H3	Leakage at matched diversity	1.73	0.95 to 2.51	0.02	supported
H4	Own-context route (transplants)	0.42	−0.35 to 1.18	0.23	not supported
H5	Gap versus recoverability (GRL)	0.06	−0.01 to 0.12	0.16	not supported

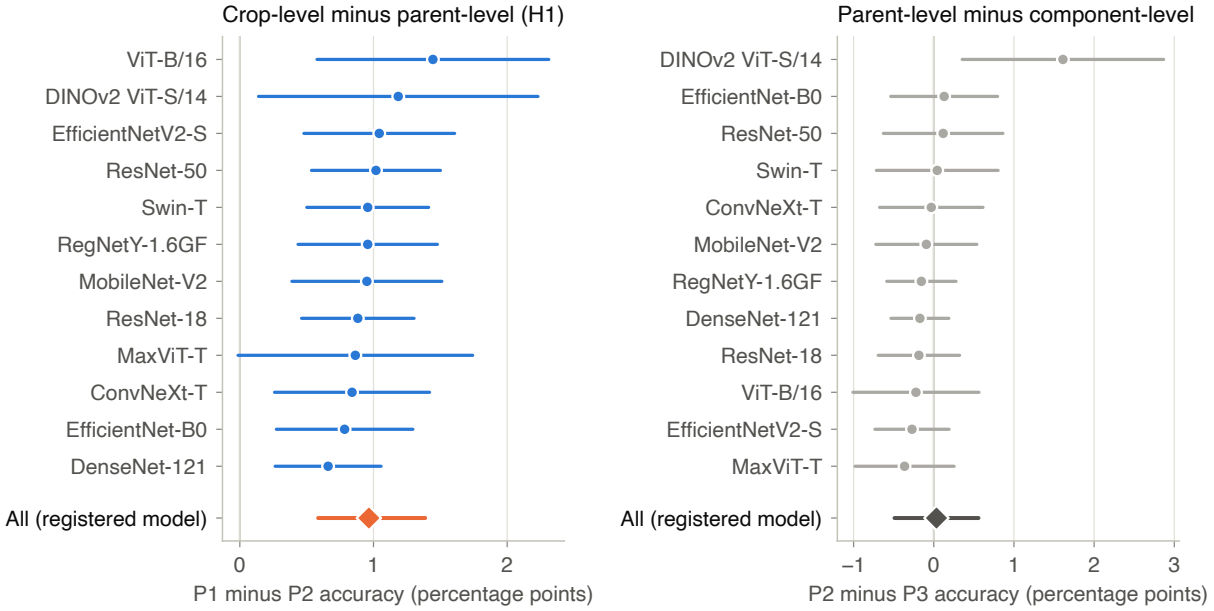


Figure 2: Left: P1 − P2 accuracy per architecture (mean and 95% t interval over 20 seeds) and the registered pooled estimate (two-way cluster bootstrap interval). Right: the same for P2 − P3, a registered secondary contrast.

6 Results

All 1,341 registered runs completed; none failed or diverged, every run was trained from the registration commit, and every result matched its registered split hash. Table 3 summarises the five registered tests.

6.1 Crop-level splitting inflates SIPaKMeD accuracy (H1)

Mean test accuracy was 97.14% under crop-level splitting (P1) and 96.18% under parent-level splitting (P2). The registered estimate of the difference is 0.97 pp (95% cluster-bootstrap interval 0.59–1.39, p <0.001; Figure 2). The mixed model gives the same estimate (Satterthwaite df = 38.8, p <0.001 with Kenward–Roger degrees of freedom), as do the seed-level sign-flip test (p <0.001) and the binomial mixed model (log odds ratio 0.304, p <0.001). All 12 architectures scored higher under P1, by 0.66–1.44 pp, and 11 of them individually had intervals excluding zero. Crop-level splitting therefore hides about a quarter of the errors: the parent-level error rate is 1.34 times the crop-level one. Macro-F1 (0.96 pp) and balanced accuracy (0.98 pp) moved by the same amount.

Merging near-duplicate-linked parents on top of parent-level splitting changed nothing (P2 − P3: 0.03 pp, −0.49 to 0.56, $p = 0.90$), so the inflation comes from shared parents, not from the residual duplicates that cross parent boundaries. The architecture ranking was largely preserved

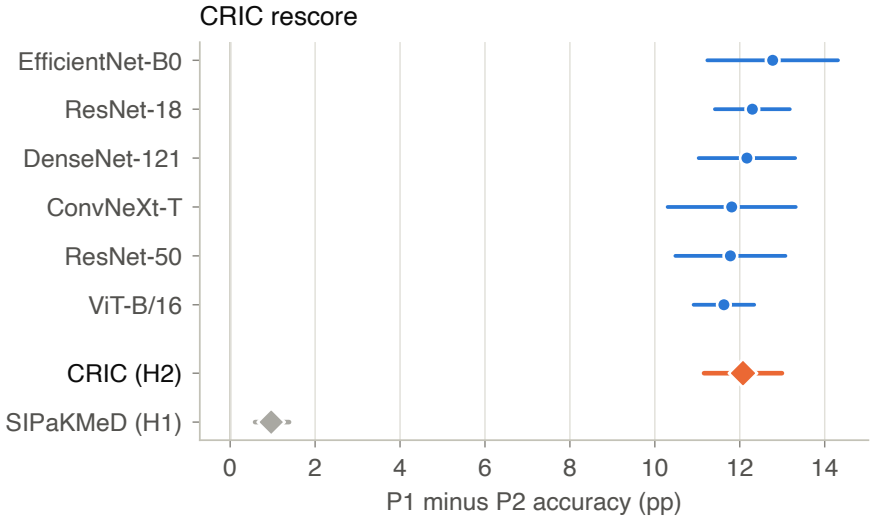


Figure 3: CRIC P1 − P2 accuracy per architecture (mean and 95% t interval over ten seeds), the registered CRIC estimate, and the SIPaKMeD estimate for scale.

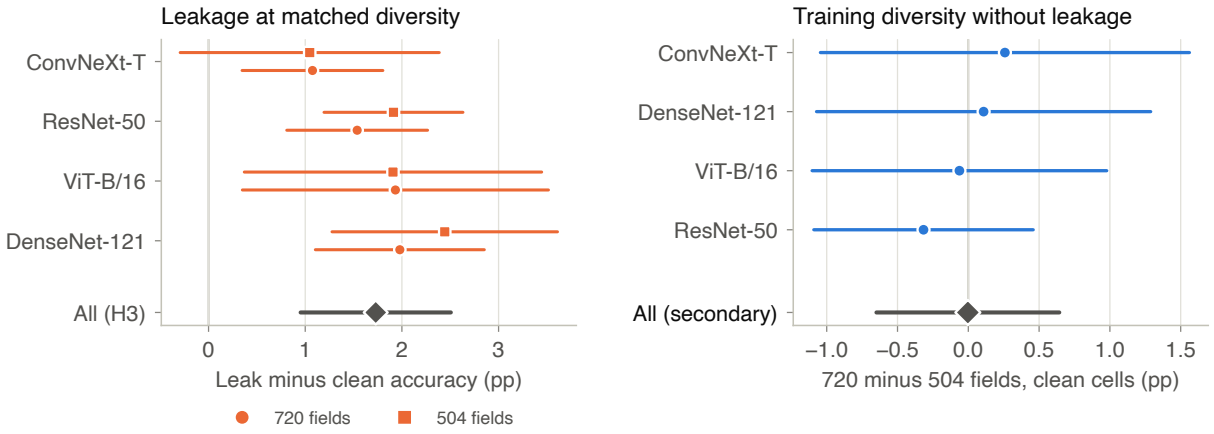


Figure 4: The 2C factorial. Left: leak minus clean accuracy at matched crop and field counts, per architecture and diversity level (mean and 95% t interval over ten seeds), and the registered H3 estimate. Right: high minus low diversity within clean cells.

(Kendall $\tau = 0.82$; 6/66 pairs reordered). Within crops tested under both protocols in different seeds (3,664 crops), accuracy was 1.29 pp higher under P1 (1.10–1.50). Within P1, test crops that shared source pixels with a training crop were more often correct (log odds 0.45, $p < 0.001$), as were crops with more training siblings (log odds 0.15 per log sibling, $p = 0.01$).

6.2 The CRIC inflation is an order of magnitude larger (H2)

CRIC accuracy fell from 90.48% under crop-level splitting to 78.40% under slide-level splitting: H2 = 12.08 pp (11.16–12.99; Holm $p < 0.001$; Figure 3). The error rate rose by a factor of 2.27. Because the CRIC classes are imbalanced, balanced accuracy fell further, from 78.53% to 58.89% (19.64 pp), and macro-F1 by 20.59 pp. The size-matched control P1-prime gave the same inflation as P1 (12.11 pp), so the gap is not a training-size effect. All six architectures showed gaps of similar size.

6.3 The inflation is leakage, not training diversity (H3)

With crop count, field count, and allocation rule held equal, adding the test fields' sibling crops to the training set raised accuracy on the common test set by 1.73 pp (0.95–2.51; Holm $p = 0.02$;

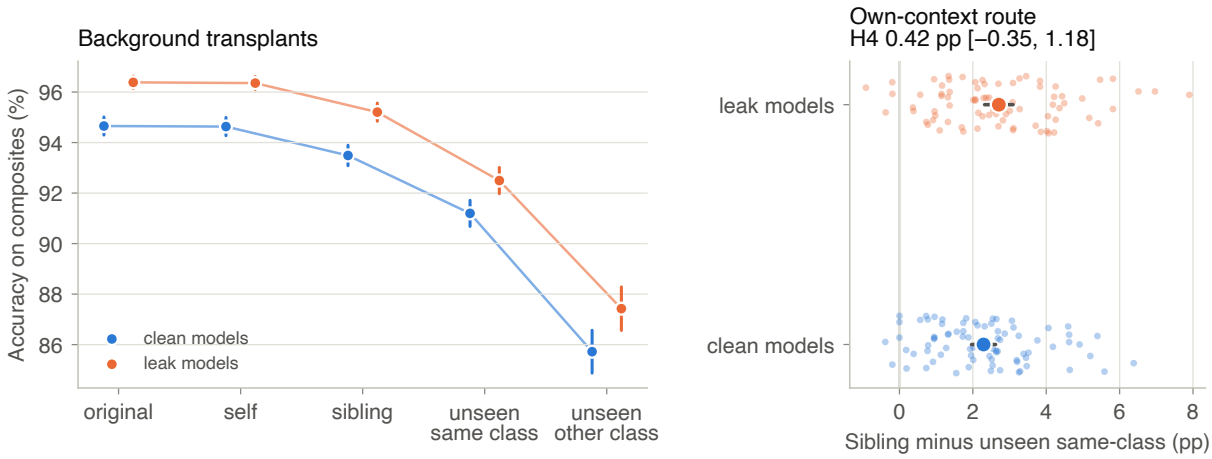


Figure 5: Left: accuracy of leak and clean 2C models on composites of their common test set (mean and 95% t interval over 80 runs each). Right: the per-run own-context route, sibling minus unseen same-class background, and the registered H4 contrast.

Figure 4). Every architecture showed the effect at both diversity levels. Increasing the number of distinct training fields by a factor of 1.43 without leakage did nothing (0.00 pp, −0.65 to 0.64), and the leakage effect did not depend on diversity (interaction −0.20 pp, $p = 0.60$). The approximate decomposition registered for the natural splits accordingly assigns the Phase 1 gap entirely to leakage (diversity component 0.00 pp).

6.4 Background transplants do not isolate the route (H4)

Compositing a cell onto its own inpainted background cost nothing, so seam and inpainting artifacts were negligible (Figure 5). Leak models were 1.73 pp more accurate than clean models on unmodified test cells and kept most of that advantage when the background was replaced: 1.72 pp on sibling backgrounds, 1.30 pp on backgrounds from unseen fields of the same class, and 1.70 pp on backgrounds of other classes. The registered own-context route was 0.42 pp larger in leak models (−0.35 to 1.18; Holm $p = 0.23$), not supported; its point estimate would assign 24% of the inflation to the background. Two observations qualify the design. Sibling backgrounds helped even clean models, which had never seen those fields, by 2.29 pp, presumably because a background from the cell's own field matches its stain; this effect is common to both arms and inflates the variance of the contrast. And backgrounds do carry class information: an other-class background moved 3.09 pp of probability toward its class (1.04–5.15, secondary).

6.5 Gradient reversal was evaded (H5)

For ConvNeXt-T the adversary was evaded: as the reversal strength grew, an independent linear probe identified the field *better* (within-class recoverability 29.5% at $\lambda = 0$, 55.1% at $\lambda = 3$), so this architecture failed the registered manipulation check and was excluded (Figure 6). For ResNet-50 recoverability fell from 24.7% to 17.6%, passing the check, and the mean gap fell from 1.17 pp at $\lambda = 0$ to between 0.28 and 0.70 pp at every $\lambda > 0$, while P2 accuracy stayed within 1 pp. The registered slope of gap on recoverability was positive but not significant (0.06 pp per pp, −0.01 to 0.12; Holm $p = 0.16$).

7 Follow-up: where does the leaked signal live?

Study 2 established *that* crop-level splitting inflates accuracy and that the inflation is leakage, but neither registered mechanism test isolated *how* the models use it: the background route

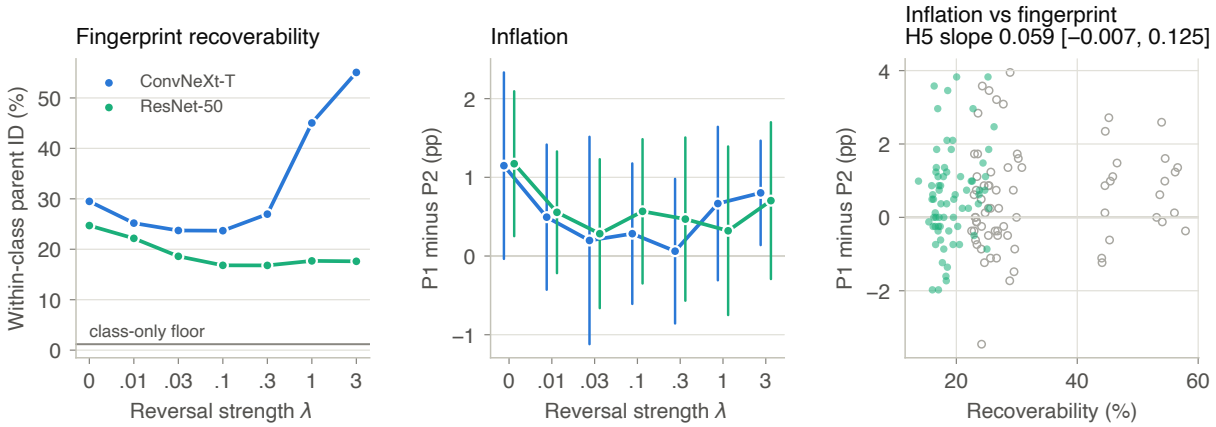


Figure 6: The gradient-reversal sweep. Left: within-class parent recoverability of the frozen features on the probe split. Middle: P1 − P2 accuracy (mean and 95% t interval over ten seeds). Right: per-seed gap against recoverability; hollow grey points are excluded from H5 (ConvNeXt-T failed the registered manipulation check).

was small and uncertain, and the adversary was evaded for one of its two architectures. We therefore registered eight further hypotheses (F1–F8) as Amendment 8 (commit `80171fe`, 2026-10-19 14:02 IST), after the Study 2 analysis and before any follow-up model was trained or any follow-up analysis was run on existing outputs. They form their own Holm family and do not alter the Study 2 verdicts. Three arms reuse saved Study 2 artifacts; three train new models on seeds no earlier arm used.

Slide context in CRIC (F1). CRIC slide images contain several classes, so a test cell's training siblings can carry a different label. For every cell, the *context class* is the majority class of the other cells of its slide. If crop-level models apply a slide-level label prior, a *discordant* cell (whose class differs from its context class) should receive more probability on the context class when its slide was in training (P1) than when it was not (P2). F1 is that within-cell difference, with a cluster bootstrap over slides.

Carrier ablations at test time (F2). Every 2C model is re-evaluated on its test set after one of five interventions that keep cell morphology: greyscale, Reinhard stain transfer to an unseen field of the same class, low-pass filtering, illumination flattening, and replacement of the background by a constant. F2 asks whether stain transfer shrinks the leak-minus-clean advantage. A quantitative Grad-CAM analysis [36] of the attribution falling outside the cell mask is reported descriptively.

Carrier removal in training (F3, F4). The 2C factorial is rerun on ten fresh seeds under three conditions: the standard recipe, every crop's background masked in training and testing, and strong optical-density (HED) colour augmentation [37] during training. F3 asks whether leakage survives with no background at all; F4 whether colour augmentation shrinks it. The standard condition doubles as an independent replication of H3.

Three ways of removing the fingerprint (F5–F8). Each manipulation lowers the within-class parent recoverability R by a different route, and the shortcut account predicts that the P1 − P2 gap falls with it. F5 erases, from every saved Phase 1 model, the top-r directions of within-class parent identity with a rank-limited LEACE projection [38] and refits the linear head ($r \in \{0, 4, 16, 64, 256\}$). F6 normalises every crop's colour statistics toward one reference at increasing strength (Reinhard). F7 repeats the adversarial arm with a probe-matched adversary

Table 4: Registered follow-up tests (Amendment 8; one Holm family). Every estimate is oriented so that the predicted direction is positive: F1 in pp of context-class probability, F2–F6 in pp of accuracy or gap, F7–F8 in pp of gap per pp of recoverability.

	Question	Estimate	95% CI	Holm p	Verdict
F1	CRIC: pull toward the slide's majority class	−4.29	−8.60 to −1.37	0.001	not supported
F2	Stain transfer shrinks the leak effect	1.27	0.38 to 2.16	0.03	supported
F3	Leak effect with backgrounds masked	1.86	1.42 to 2.30	<0.001	supported
F4	Colour augmentation shrinks the leak effect	−0.23	−0.81 to 0.35	1.00	not supported
F5	LEACE erasure shrinks the gap	−0.01	−0.10 to 0.07	1.00	not supported
F6	Stain normalisation shrinks the gap	0.02	−0.41 to 0.44	1.00	not supported
F7	Gap tracks R (linear adversary)	0.000	−0.038 to 0.039	1.00	not supported
F8	Gap tracks R (pooled)	0.000	−0.005 to 0.005	1.00	not supported

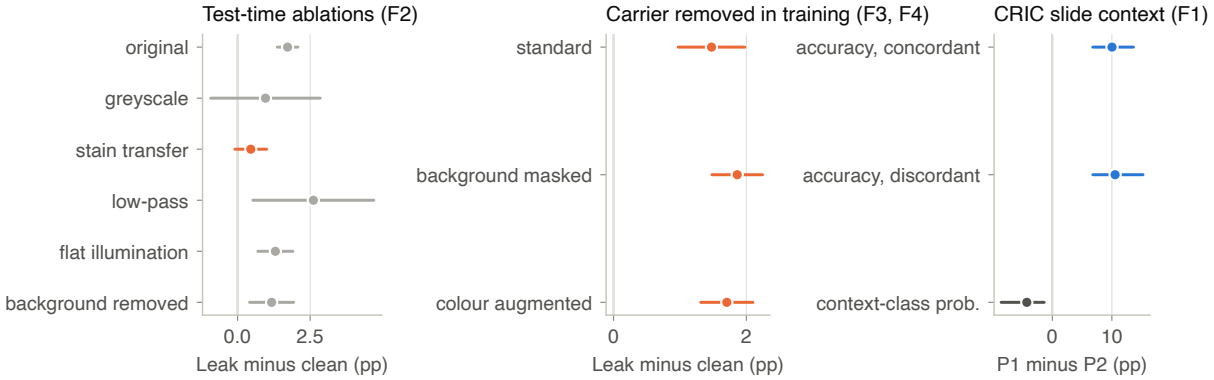


Figure 7: Where the leaked signal lives. Left: leak minus clean accuracy of the 160 Study 2 2C models under each test-time ablation (mean and 95% t interval over 80 architecture–seed–diversity pairs); stain transfer (F2) highlighted. Middle: the same contrast for models trained on fresh seeds under the standard recipe, with every background masked (F3), or with HED colour augmentation (F4). Right: CRIC, within-cell P1 − P2 accuracy for cells whose class agrees or disagrees with their slide's majority, and the context-class probability for disagreeing cells (F1); slide-bootstrap intervals.

(standardised features, one linear layer, periodic re-initialisation) that cannot hide parent identity from a linear probe while leaving it readable. F8 pools the dose-response over all three. The H5 regime rule and manipulation check apply to F5 and F7 and decide which levels enter F8.

7.1 Follow-up results

All 1,580 follow-up tasks completed without failure or divergence. One labelling bug in the analysis preparer (standard carrier runs labelled as colour-augmented) halted the registered script before it produced any output; it was fixed and recorded as Amendment 9 before any F3 or F4 estimate was seen. Table 4 gives the eight registered tests.

The leak needs no background (F3). Trained and tested with every background replaced by a constant, leak models beat clean models by 1.86 pp (1.42–2.30; Holm p <0.001), at least as much as under the standard recipe on the same fresh seeds (1.47 pp, which also replicates H3: 1.47 pp, 0.06–2.89). At test time, removing the background from the original 2C models' inputs likewise left most of the advantage (1.17 pp of 1.72; Figure 7).

The leak is tied to the field's colour rendering at test time (F2), but colour augmentation does not prevent it (F4). Moving each test cell's LAB colour statistics to those of an unseen field of the same class cut the leak advantage from 1.72 to 0.46 pp, a reduction of 1.27 pp (0.38–2.16; Holm $p = 0.03$). The other interventions with comparable accuracy costs did

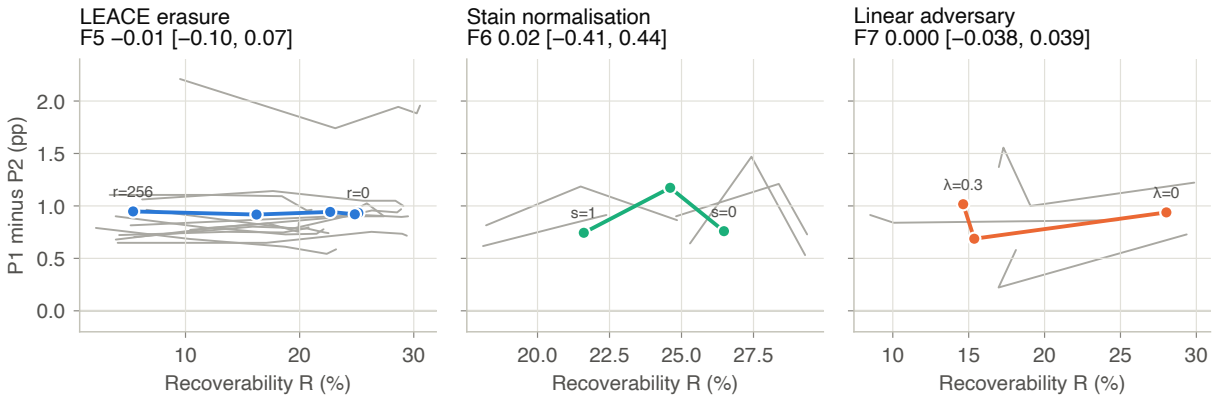


Figure 8: The P1 − P2 gap against within-class parent recoverability R along three manipulations (grey: architectures; colour: pooled over architectures that passed the manipulation check, at levels they share; only registered in-regime levels are shown). Left: rank-limited LEACE erasure, ranks 0–256, twelve architectures. Middle: Reinhard normalisation strength 0–1, four architectures (all failed the manipulation check and are excluded from F8). Right: probe-matched linear adversary, three architectures.

not: with illumination flattened the advantage was 1.30 pp and with the background removed 1.17 pp; greyscale and low-pass filtering degraded accuracy too far to be informative. Yet models trained with strong HED colour augmentation leaked as much as standard models (1.71 versus 1.47 pp; F4 reduction −0.23, −0.81 to 0.35).

Exploratory: not shared pixels. In leak cells a third or more of the test crops overlap a training sibling in the source image. The leak advantage was the same on crops that share no pixel with any training crop (1.68 pp) as on crops that do (1.78 pp), so the leak generalises to new cells from a seen field rather than reflecting memorised pixels; stain transfer removed it in both subsets (0.70 and 0.08 pp). This split was not registered.

No slide-level label prior (F1). CRIC cells whose class disagreed with the majority class of their slide received *less*, not more, probability on that majority class when their slide was in training (−4.29 pp, −8.60 to −1.37, over 1,144 cells from 230 slides; F1 not supported). Crop-level training raised accuracy as much for these cells (10.55 pp) as for cells agreeing with their slide (10.02 pp; contrast −0.52 pp, −5.75 to 4.66): seen slides help each cell toward its own class, not toward the slide's typical class.

The gap does not track the fingerprint (F5–F8). Three manipulations lowered R by different routes, and none moved the gap (Figure 8). Rank-256 LEACE erasure cut recoverability from 25.1% to 5.4% (the class-only floor is 1.2%) in all twelve architectures while costing P2 0.17 pp, and the gap stayed at 0.93 to 0.95 pp (F5 −0.01, −0.10 to 0.07). The probe-matched linear adversary passed its manipulation check in all three architectures, lowering R from 24.8–29.9% to 8.5–18.1% inside the regime, with a slope of gap on R of 0.000 (−0.038 to 0.039). Stain normalisation lowered R only from 26.5% to 21.6% while costing P2 1.54 pp, failed its manipulation check for every architecture, and left the gap unchanged (F6 0.02 pp per unit strength, −0.41 to 0.44). Pooled over the 15 included manipulation–architecture cells, the slope was 0.000 (−0.005 to 0.005): even a 20-pp loss of recoverability would move the gap by at most about 0.1 pp. The device probe used for R agreed with the registered sklearn probe on the 280 Study 2 adversarial runs ($r = 0.999$).

Grad-CAM sees no difference (descriptive). Leak and clean models placed the same share of Grad-CAM attribution outside the cell mask (21.8% on average, against a background

pixel share of 34.6%; leak minus clean −0.09 pp, −0.57 to 0.39). The maps depended on the trained head (median Spearman correlation 0.08 with a random head), so the absence of a difference is not a failure of the method's sanity check; it is the expected result if the leaked signal lies in how the cell is rendered rather than in where the model looks.

8 Discussion

Crop-level splitting inflates accuracy, and the inflation is leakage. With one registered test, sized by simulation, Study 2 detects what Study 1's six separate tests could not: on SIPaKMeD, crop-level splitting raises accuracy by 0.97 pp, consistently across twelve architectures from ResNet-18 to DINOv2. The factorial removes the obvious alternative. At equal crop and field counts, giving the model the test fields' siblings raises accuracy by 1.73 pp, and the effect replicated on fresh seeds (1.47 pp), while giving it 43% more distinct fields without leakage does nothing. The natural gap is therefore leakage, not the extra training diversity that crop-level splits also bring.

The size of the inflation follows the structure. SIPaKMeD releases about four crops per field and every field belongs to one class; CRIC releases about 29 cells per slide image, and slide identity carries most of the class information (Section 4). The inflation scales accordingly, from 0.97 pp (an error rate 1.34 times higher at the parent level) to 12.08 pp (2.27 times), with balanced accuracy on CRIC falling by 19.64 pp. The closed forms of Section 4 predict which benchmarks are exposed before any model is trained.

How the leak is used. The two registered mechanism tests of Study 2 and the eight follow-up tests converge on a description by elimination. The leaked advantage does not need a background: it survives masking every background in training and testing (F3) and removing it at test time, and the background-transplant route was small and uncertain (H4). It is not memorisation of shared pixels: it is as large for test cells that share no pixel with any training crop (exploratory). It is not a slide-level label prior: on CRIC, seen slides helped cells whose class disagreed with the slide's majority as much as cells that agreed, and pulled them *away* from the majority class (F1). And it is not carried by a separable, linearly decodable code for the field: erasing that code from the features in closed form, or suppressing it with an adversary that cannot hide it from a linear probe, cut recoverability by 38–91% of its baseline without moving the gap (F5, F7, F8). What remains is that the advantage depends on the cell being rendered as its own field renders cells: transferring a test cell's colour statistics to another field of the same class removed about three quarters of it (F2). The simplest account consistent with all of this is that a model trained on a field's siblings calibrates its class evidence to that field's appearance (stain, colour balance, and the texture they produce in the cell itself), so cells from a seen field sit further inside their class region. Random colour jitter during training did not prevent this (F4), presumably because jitter perturbs each image around its own appearance and leaves the field-specific structure of that appearance learnable.

Decodable is not used, and saliency is not use either. Study 1 established *availability*: a probe recognised fields at 109 times chance. Study 2 shows that availability and use come apart. The field identity a linear probe can read can be removed almost entirely without touching the inflation, in line with evidence from language models that probe-decodable information need not be what a model relies on [39, 40]. Grad-CAM, the tool most often used to argue that a medical classifier looks at the right thing, showed leak and clean models attending to the same places; a saliency audit would have cleared both. Neither probes nor saliency maps can certify that an evaluation is free of hierarchical leakage; only the split can.

What survives, and what does not fix it. Within a benchmark, crop-level splitting preserved the architecture ranking reasonably well (Kendall $\tau = 0.82$), so relative comparisons under one protocol are not meaningless. Absolute numbers are another matter: results reported for the same dataset under crop-level and parent-level splits are not comparable, and on CRIC the protocol difference is several times larger than the spread between the six architectures under either protocol. Neither of the usual robustness measures removed the inflation: background removal left it intact (F3), strong colour augmentation left it intact (F4), and stain normalisation reduced recoverability only slightly while costing accuracy (F6).

8.1 Limitations

Parents are not patients. SIPaKMeD fields, Cx22 source images, and CRIC slide images are the deepest units the releases identify. Several fields can come from one patient, so parent-disjoint evaluation is necessary but may not be sufficient for clinical independence, and every estimate here is a lower bound on what patient-level grouping would remove.

The factorial trains on fewer crops. Holding crop and field counts equal across leak and clean cells requires 220 crops per class (1,100 in total, against 3,239 in Phase 1), so the 2C and 2B effects are measured on smaller training sets than the headline contrast. The decomposition that transfers them to the natural splits assumes the leakage effect scales with the leak rate and the diversity effect scales with the logarithm of the field-count ratio; it is labelled approximate.

Transplants are imperfect interventions. The foreground mask is a fixed threshold rule, inpainting leaves a smoothed background, and feathering mixes a few pixels of donor and recipient. The *self* condition measures these artifacts, and every contrast compares conditions that share them, but composites are still off the training manifold.

Suppression and erasure act on pooled features. The adversaries and the LEACE eraser remove parent identity that is linearly readable from the pooled penultimate features. Identity carried non-linearly, or in spatial features before pooling, can survive; the null dose-response therefore rules out a linearly decodable field code as the carrier, not every form of field information. The colour arms used one augmentation strength and one normalisation reference, and stain normalisation failed its own manipulation check.

The follow-up was registered after Study 2. Amendment 8 was written knowing the Study 2 results and was motivated by them. Its hypotheses, designs, and analyses were fixed before any follow-up model was trained or any follow-up analysis was run, but they are follow-up questions, and the one analysis added afterwards (the pixel-overlap split) is labelled exploratory.

One recipe. Every model uses the same fixed fine-tuning recipe. Stronger augmentation, longer training, or stain normalisation could change both the size of the inflation and the models' reliance on acquisition context.

One host. All runs used one four-GPU server with a four-core CPU allowance, which is why augmentation runs on the GPU. Wall-clock times are not portable, and the GPU implementation of antialiased resizing differs from PIL's by uint8 rounding.

No clinical claim. The study audits research benchmarks. It does not estimate screening sensitivity, specificity, prevalence-calibrated performance, or effects on readers.

8.2 Recommendations for dataset releases and evaluations

1. **Release the hierarchy.** Every observation should carry a stable observation identifier, its acquisition parent (field, source image, or slide), a patient or specimen identifier where ethics and law permit, and, if duplicates were audited, a duplicate-component identifier.

2. **Split at the deepest released unit and publish the split.** Evaluate on parent-disjoint (or patient-disjoint) folds and release the exact fold manifest with its hash, so that results reported under one dataset name are comparable.

3. **Report both units.** State observation and parent counts for every split, and the fraction of test observations with a training sibling.

4. **Check the structure before training.** The closed forms of Section 4 (`cytofair risk-report`) give the expected straddle, sibling rate, diversity ratio, and class–parent dependence of a crop-level split from the parent table alone.

5. **Treat one question as one test.** When the same question is asked of many architectures, test it once with a mixed model that treats architectures and splits as random effects, and size the study by simulation; report per-architecture contrasts descriptively.

6. **Do not certify a split with a probe, a saliency map, or an augmentation.** Leakage inflated accuracy with field identity erased from the features, with identical Grad-CAM maps, with backgrounds removed, and with strong colour augmentation. Only a parent-disjoint split removes it.

7. **Keep the evidence.** Save per-sample predictions, final weights, and the exact split for every run, so that confusion matrices, calibration, and later mechanistic analyses do not require retraining.

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A Post hoc omnibus reanalysis of Study 1

Fitting model (1) to the 60 Study 1 P1 and P2 runs (six architectures, five seeds) gives a P1–P2 difference of 1.37 pp (SE 0.38, Satterthwaite $p = 0.005$, df $= 9.1$). This analysis was specified after the Study 1 data were seen; it motivated Study 2 and supplied its pilot variance components, and it is not confirmatory.

B Power analysis

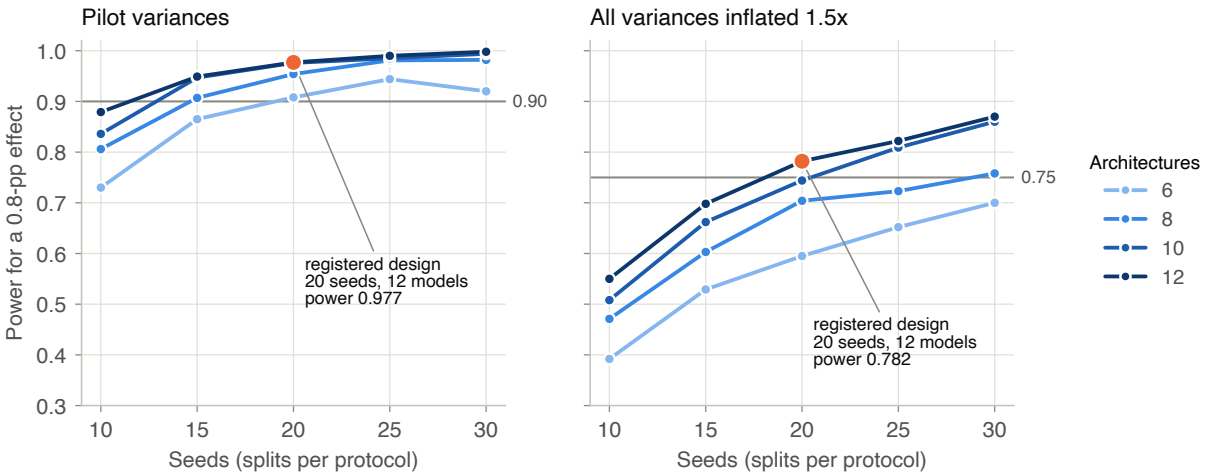


Figure 9: Simulated power of model (1) for a 0.8-pp effect (1,000 simulations per design). The registered design (twenty seeds, twelve architectures) reaches 0.977 with pilot variances and 0.782 with every variance inflated by half.